SCSR2043 - OPERATING SYSTEMS (Lab Tutorial 0)

Written by: Dr. Farkhana Muchtar

LAB 0: INSTALLATION EXERCISE

Installing and Running Linux OS in Windows 10 Using Oracle VirtualBox

1 Installation Steps

1.1 Windows OS Command Line Environment

We will use PowerShell 7 (an open-source tool) and Windows Terminal to improve the Windows command line, making it easier to use and more visually appealing than the default options.

Windows 10/11 includes two command line tools by default: Command Prompt (an older tool from DOS, Windows 95, and Windows NT) and Windows PowerShell (more powerful but not very attractive).

1.1.1 Installing Open Source PowerShell (PowerShell 7)

Step 1: Download PowerShell 7 Installer from this link

https://github.com/PowerShell/PowerShell/releases/tag/v7.5.3

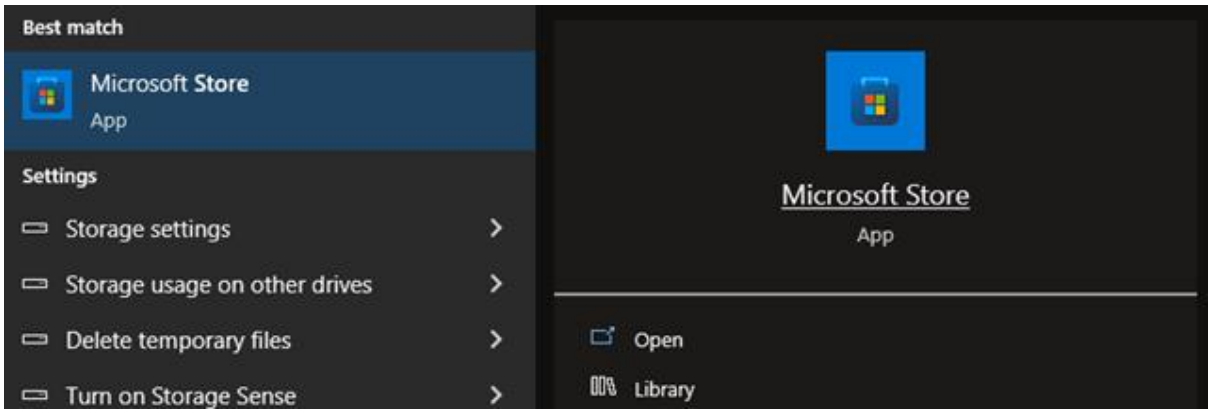
Step 2: Scroll down until the Assets section. Choose **PowerShell-7.5.3-win-x64.msi** file.
Step 3: After download, simply double click the installer file

1.1.2 Installing Windows Terminal

Step 1: Search **Store** at Windows Search



Step 2: Click **Microsoft Store**

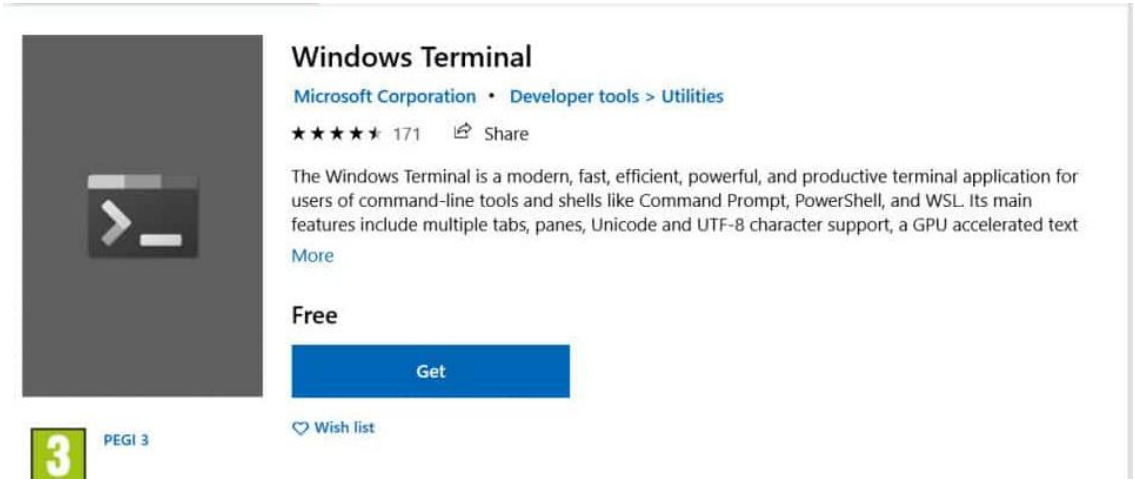


Step 3: In Microsoft Store, search for **Windows Terminal**

Step 4: Simply click **Get** button to install **Windows Terminal**.

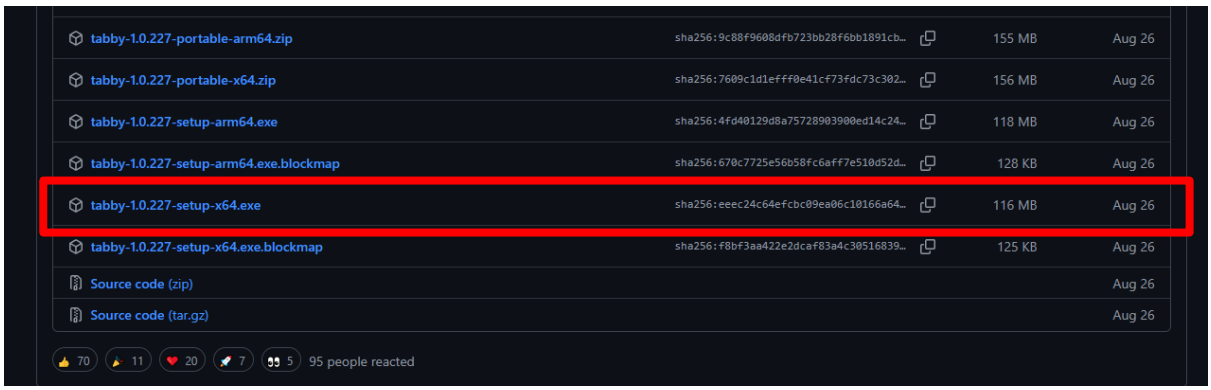
SCSR2043 - OPERATING SYSTEMS (Lab Tutorial 0)

Written by: Dr. Farkhana Muchtar



If you have trouble installing Windows Terminal from the Microsoft Store, try Tabby as an alternative terminal emulator for Windows 10/11.

Step 1: Go to https://tabby.sh/
Step 2: Find download button and click the button to go to the github download page.
Step 3: Locate the correct setup file and click it to download it to your computer.
Step 4: Click the downloaded file to install **tabby**.



Tabby might not as user friendly as **Windows Terminal** but it has more features that you can explore.

* For Mac OS user, it is enough to use default **Terminal** app in your Mac OS. Just search **Terminal** in Finder to start using **Terminal** in your Mac OS.

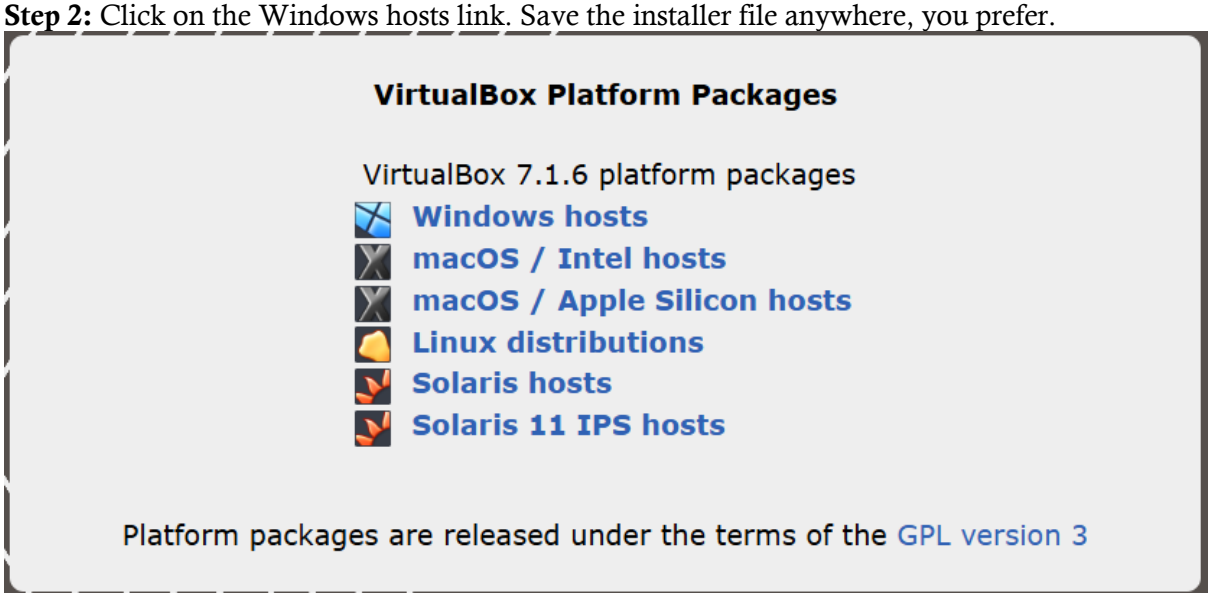
Written by: Dr. Farkhana Muchtar

1.2 Installing Virtualbox as Virtualization Solution

VirtualBox is a free Type 2 hypervisor that lets you run virtual machines quickly and reliably, depending on your computer's hardware. Even though it is entirely free, VirtualBox still provides features that many other hypervisors either do not have or need a premium subscription to use.

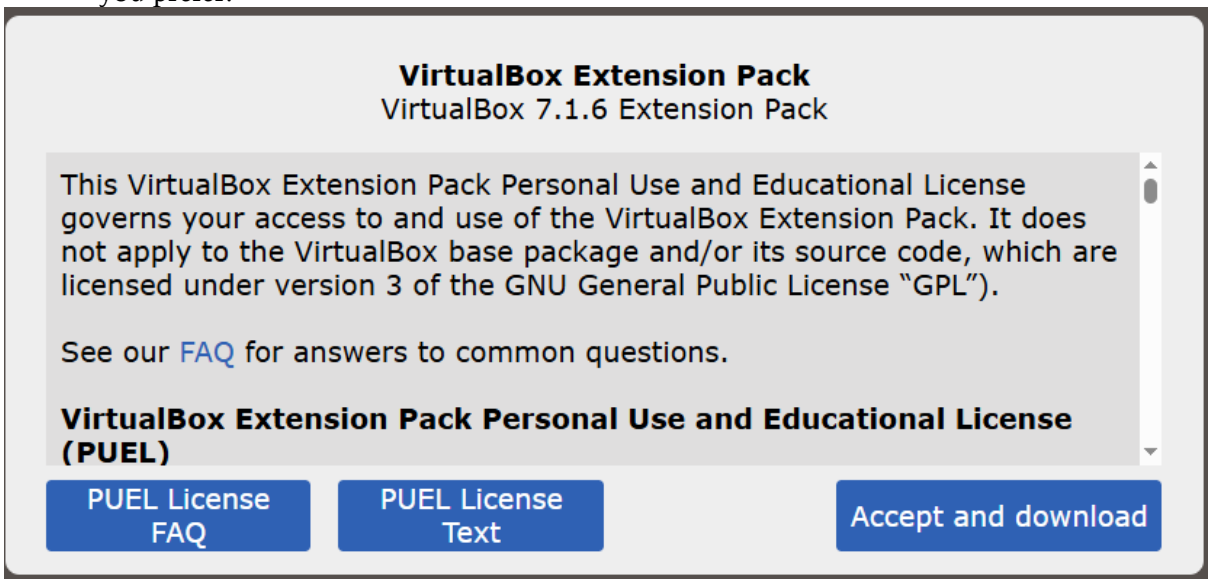
Step 1: Go to the Oracle VirtualBox download page using the below link.
https://www.virtualbox.org/wiki/Downloads

Step 2: Click on the Windows hosts link. Save the installer file anywhere, you prefer.



Step 3: Navigate to the installer file and double-click on it for the installation process. Stick with the default settings during the installation process.

Step 4: At the same website page, under the '**VirtualBox Extension Pack**' section, click the '**Accept and download**' button to download the latest pack. Save the installer file anywhere, you prefer.



Step 5: Navigate to the installer file and double-click on it for the installation process. Now the Virtualbox already installed in your computer/laptop.

Written by: Dr. Farkhana Muchtar

1.3 Steps to Install Ubuntu 24.04 LTS Minimal Cloud Image on Virtual Box

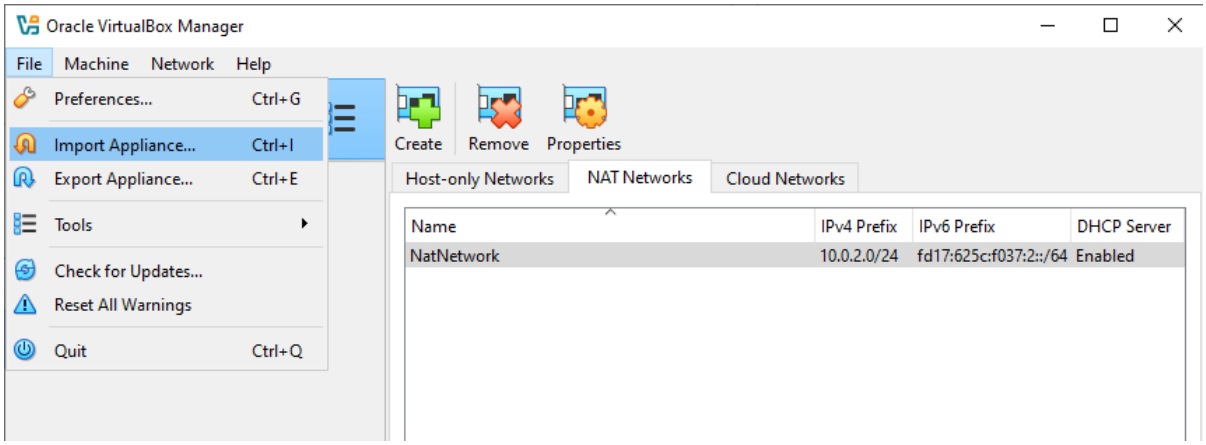
Step 1: Download Ubuntu 24.04 LTS Cloud Image
Since we're installing Ubuntu 24.04 Noble Numbat using VirtualBox, you can download a ready-made file (called an appliance) from the Ubuntu 24.04 CD Image page.

Visit this link
https://cloud-images.ubuntu.com/noble/current/

and on the page, download **noble-server-cloudimg-amd64.ova** file.

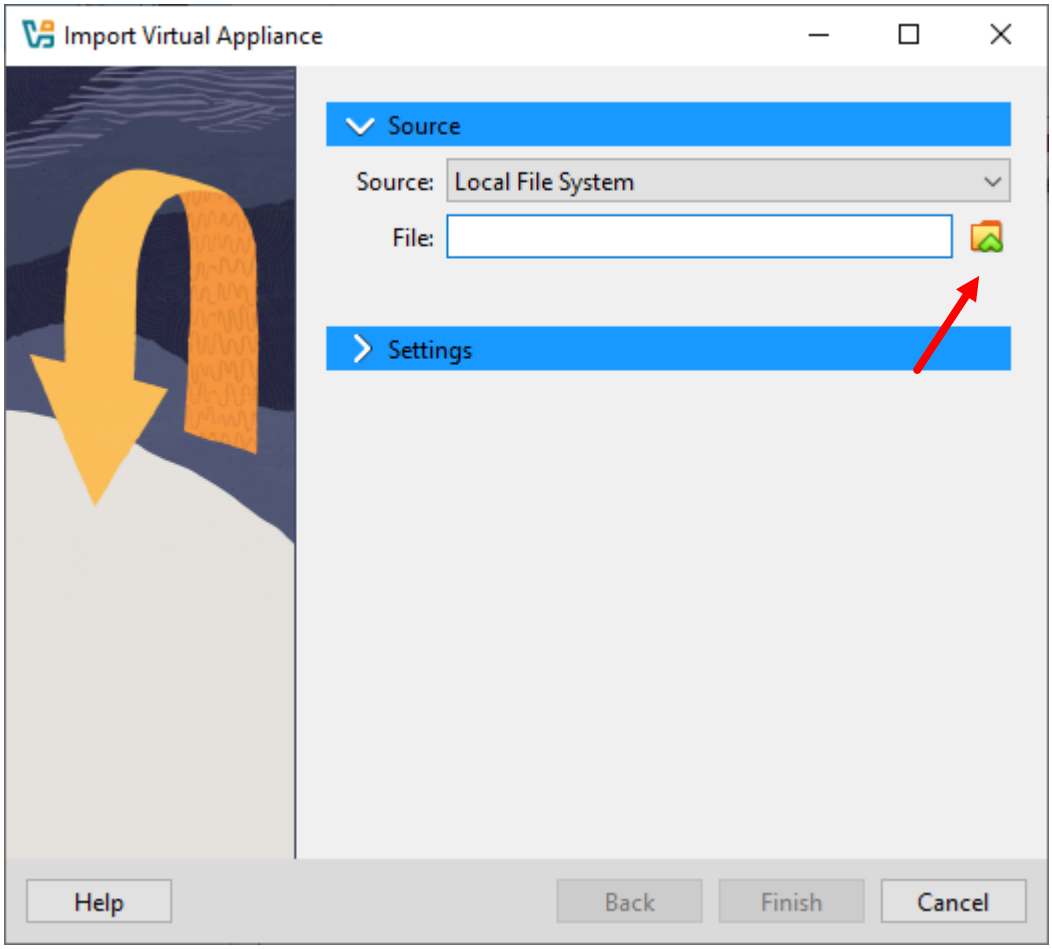


Step 2: Import Appliance in VirtualBox
Open VirtualBox on your computer. Then, go to the **File** menu and choose **'Import Appliance…'**.

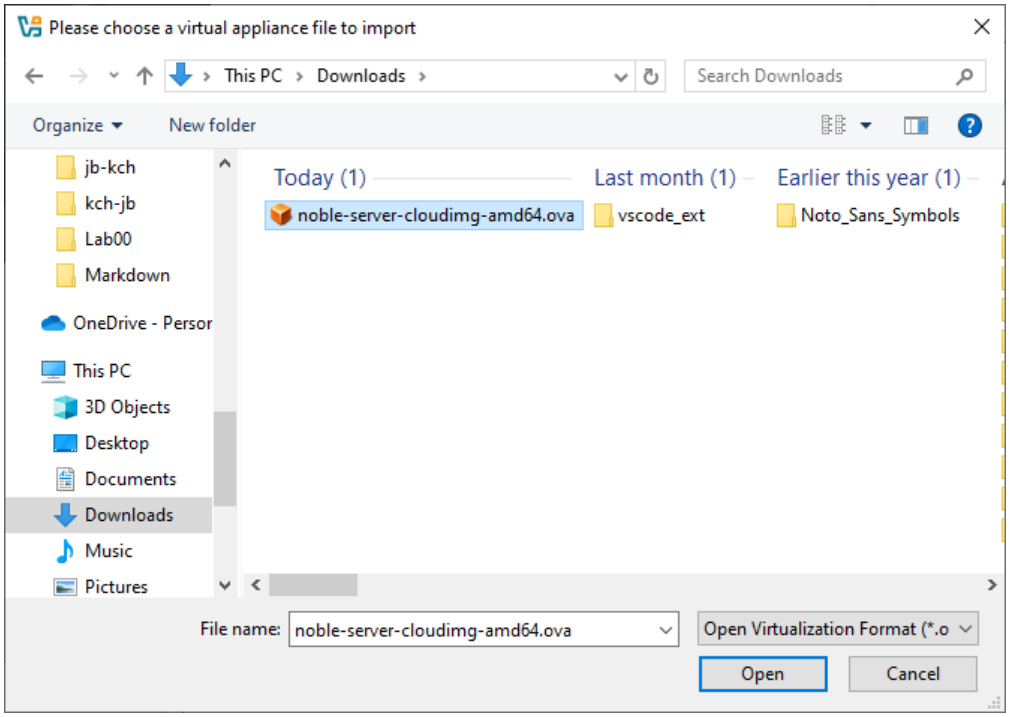


Written by: Dr. Farkhana Muchtar

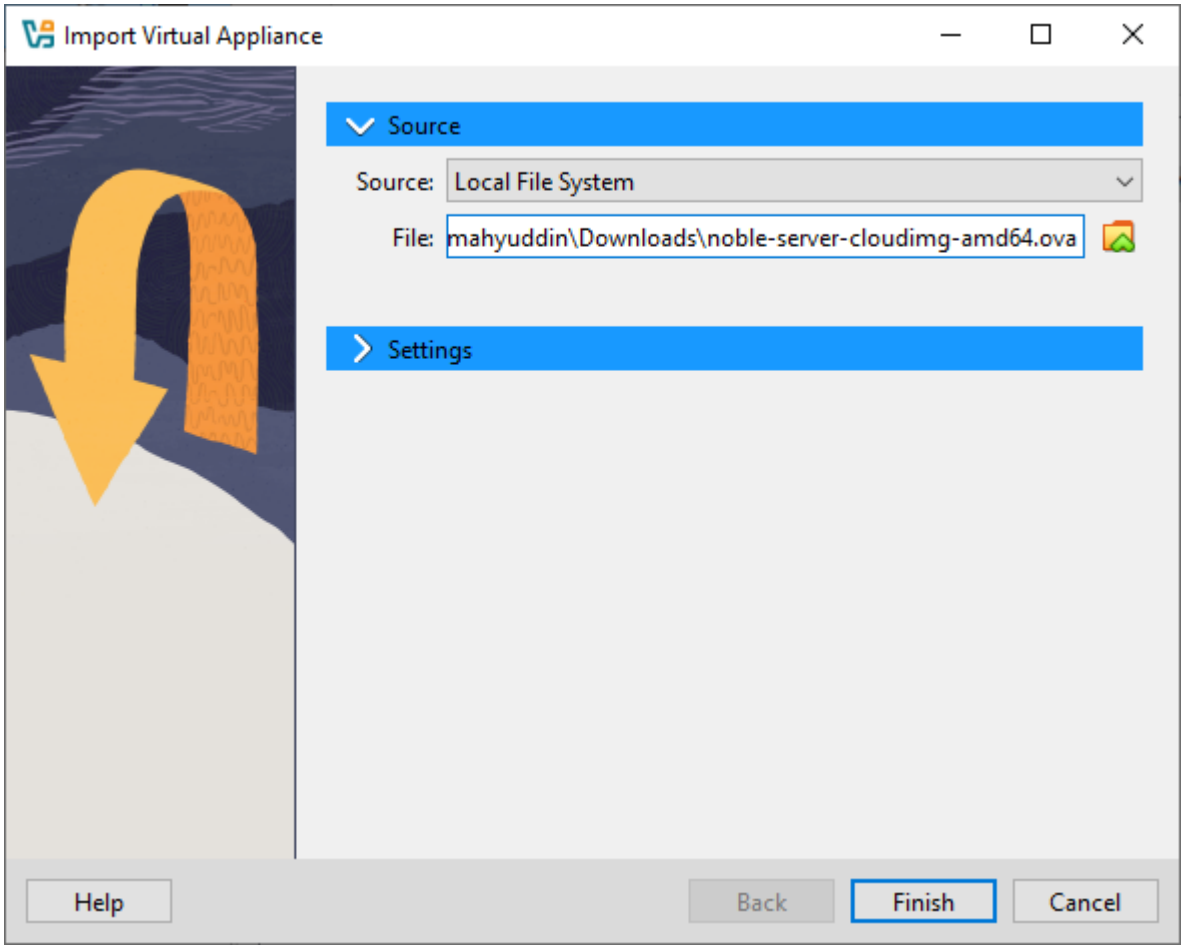
Click the folder icon



and locate the Ubuntu 24.04 Noble Numbat OVA file you downloaded. Then, click the 'Open' button.

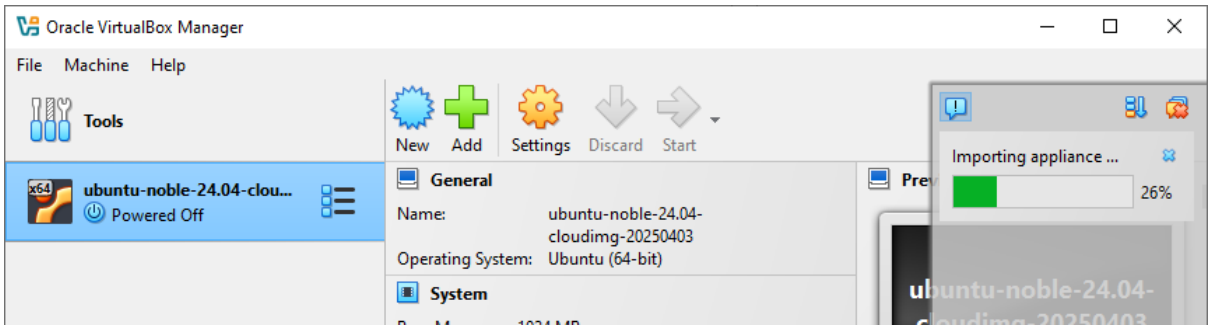


Written by: Dr. Farkhana Muchtar



After that, simply hit the **Finish** button.

Virtualbox will import the appliance image and creating a new virtual machine based on that image.

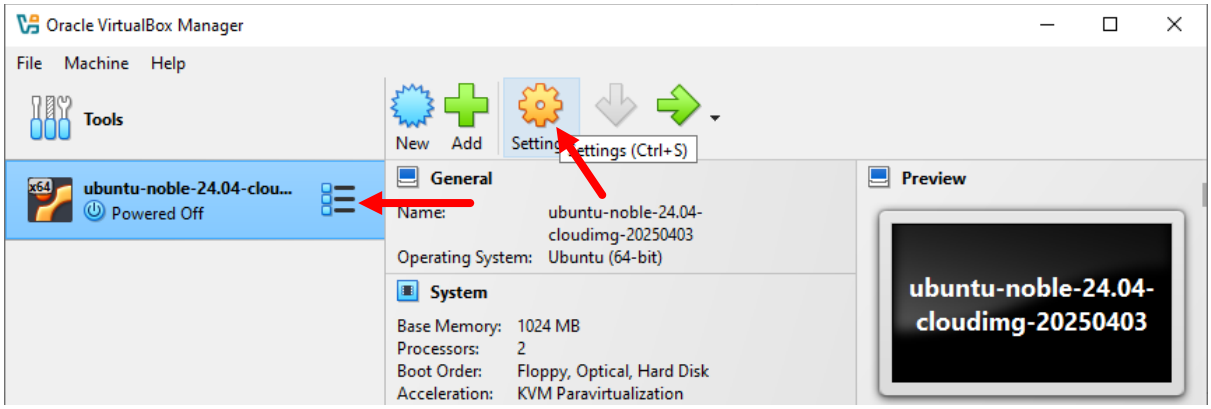


SCSR2043 - OPERATING SYSTEMS (Lab Tutorial 0)

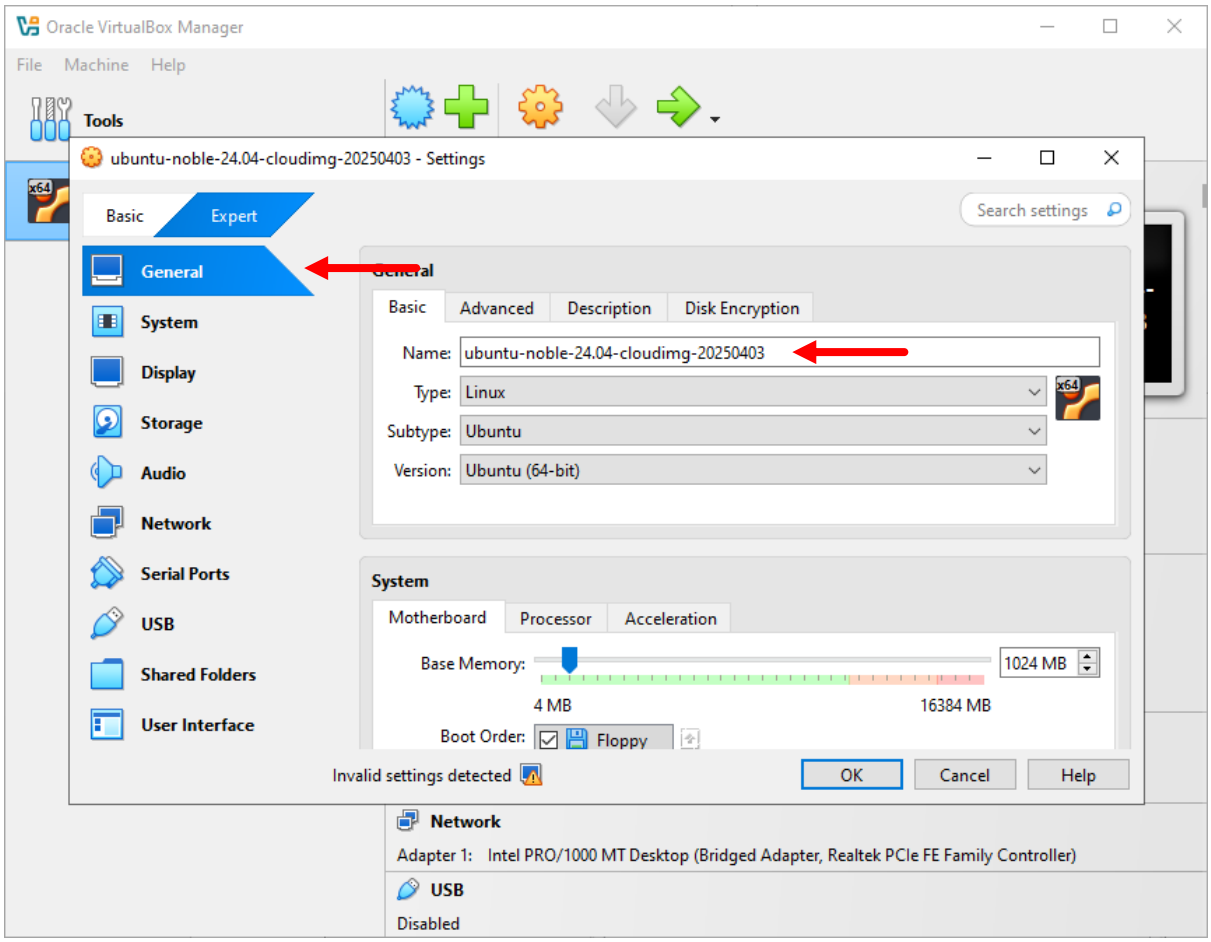
Written by: Dr. Farkhana Muchtar

Step 3: Setting up the virtual machine

Select the virtual machine and click Setting icon to change the setting of the virtual machine.

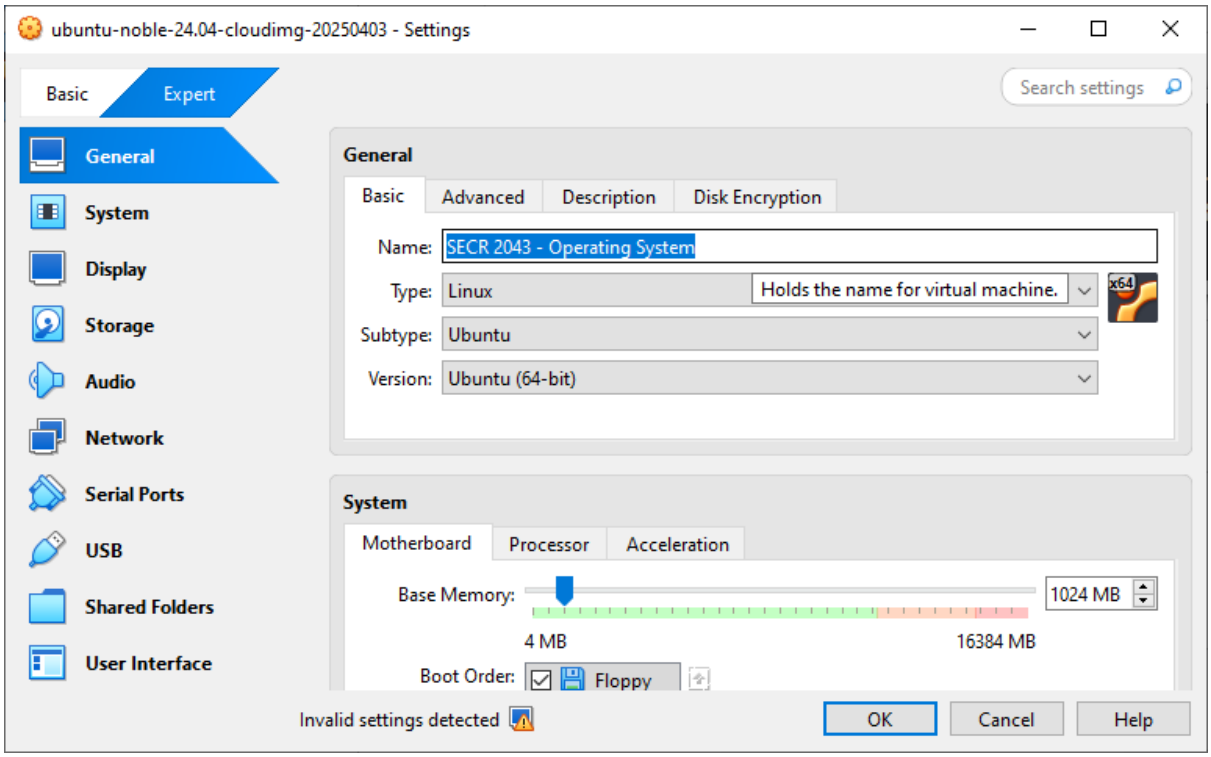


First, you need to set a name for your new virtual machine. By default, you will be to setting up **General** configuration of the virtual machine.

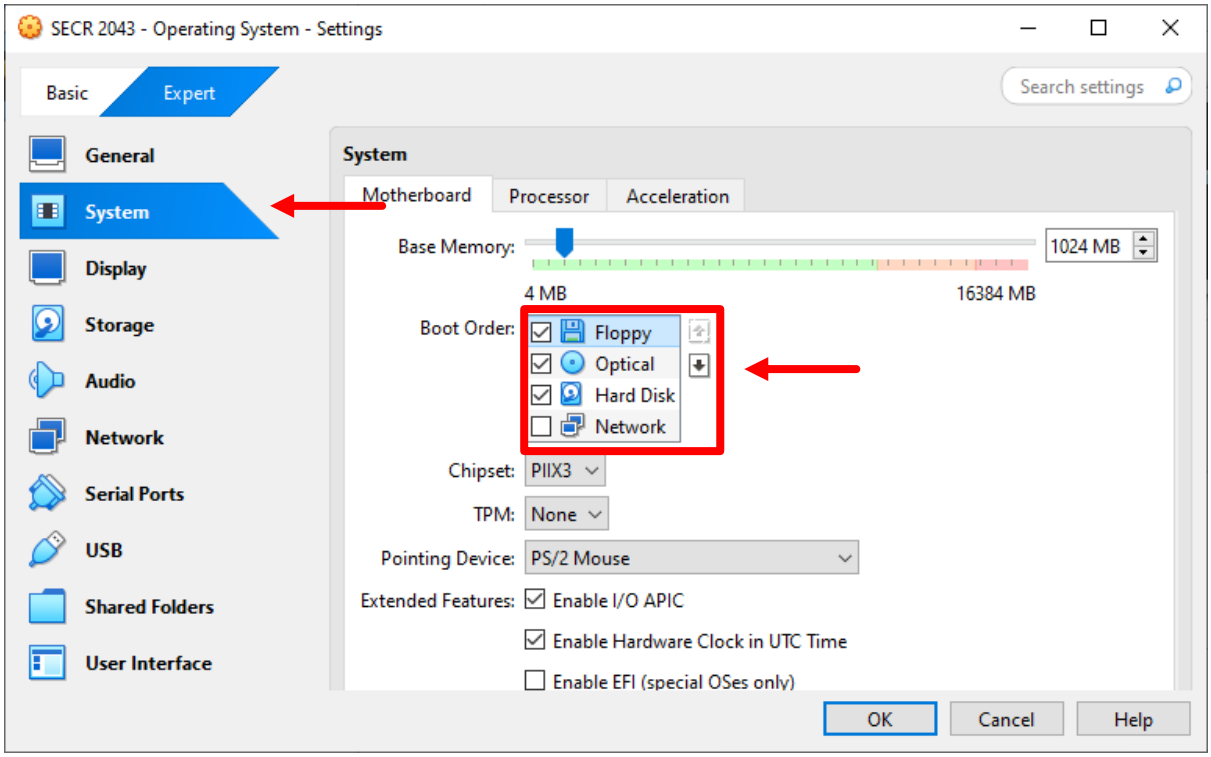


Written by: Dr. Farkhana Muchtar

In these settings, we'll simply give our virtual machine a more descriptive name. For example, **SECR 2043 – Operating System**.

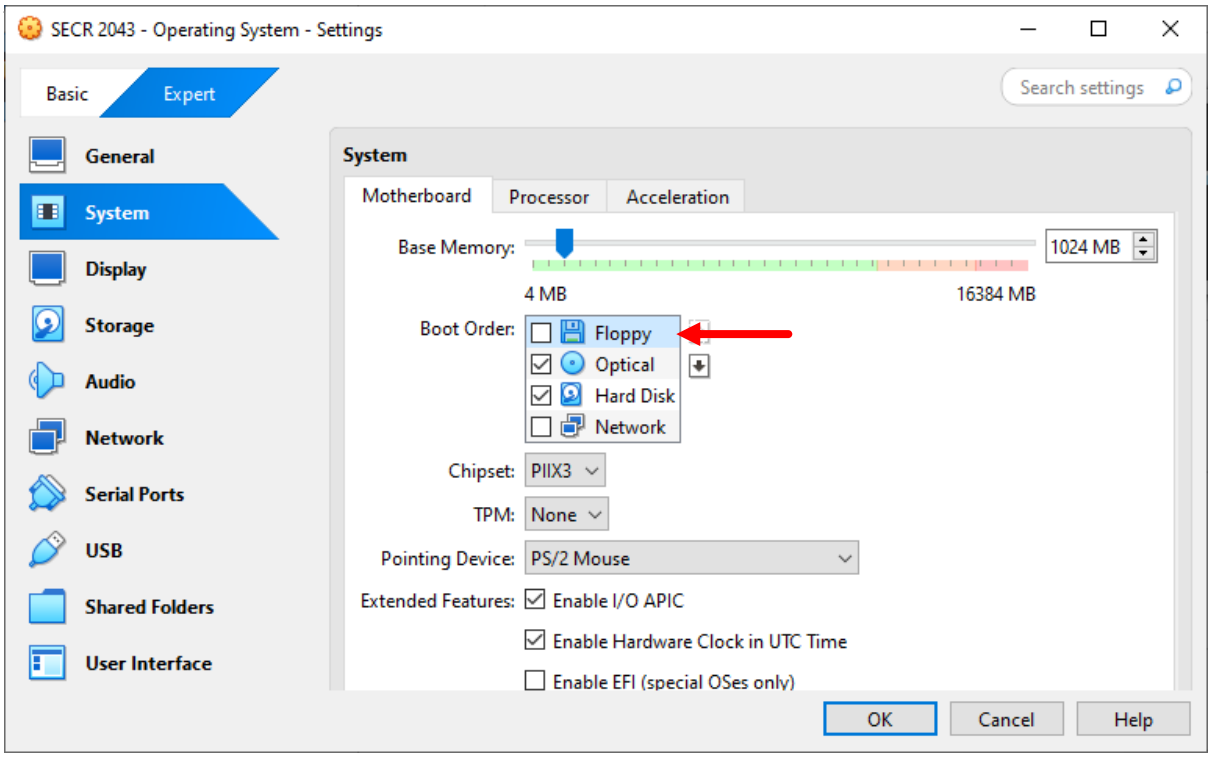


After that, we will do some **System** setting, specifically on **Boot Order** of our virtual machine.

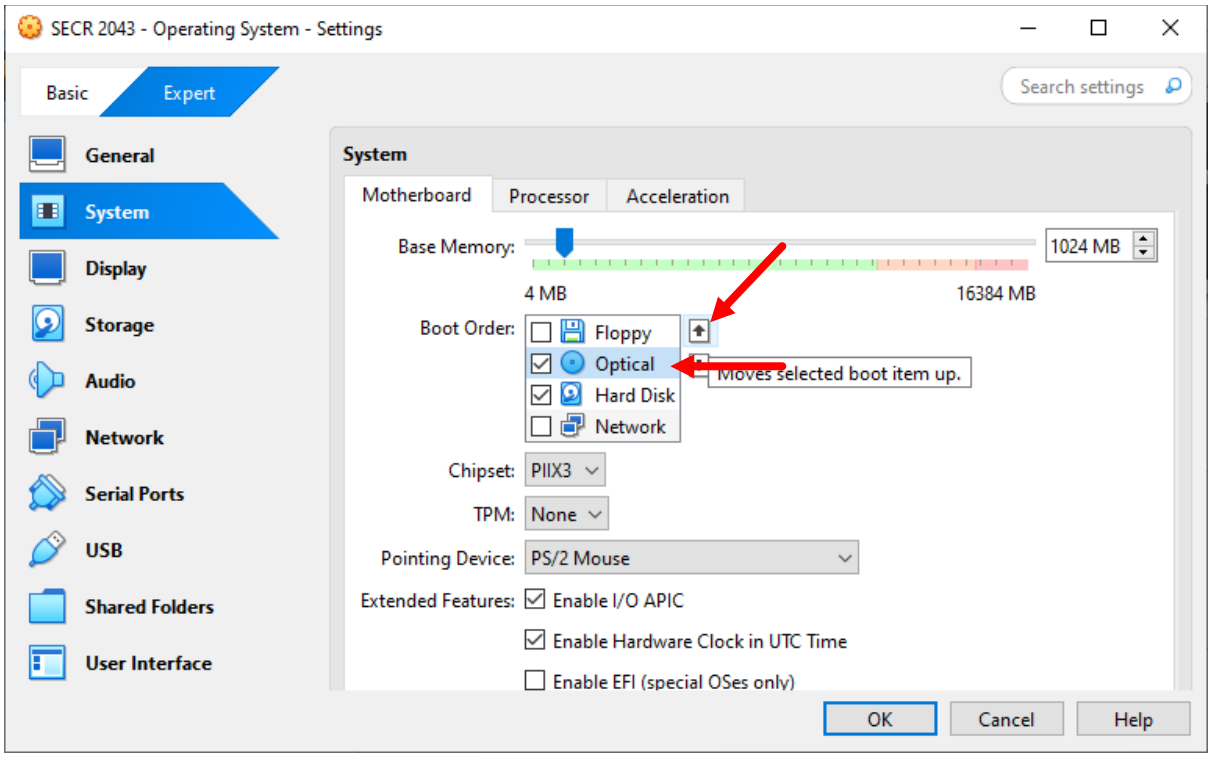


Written by: Dr. Farkhana Muchtar

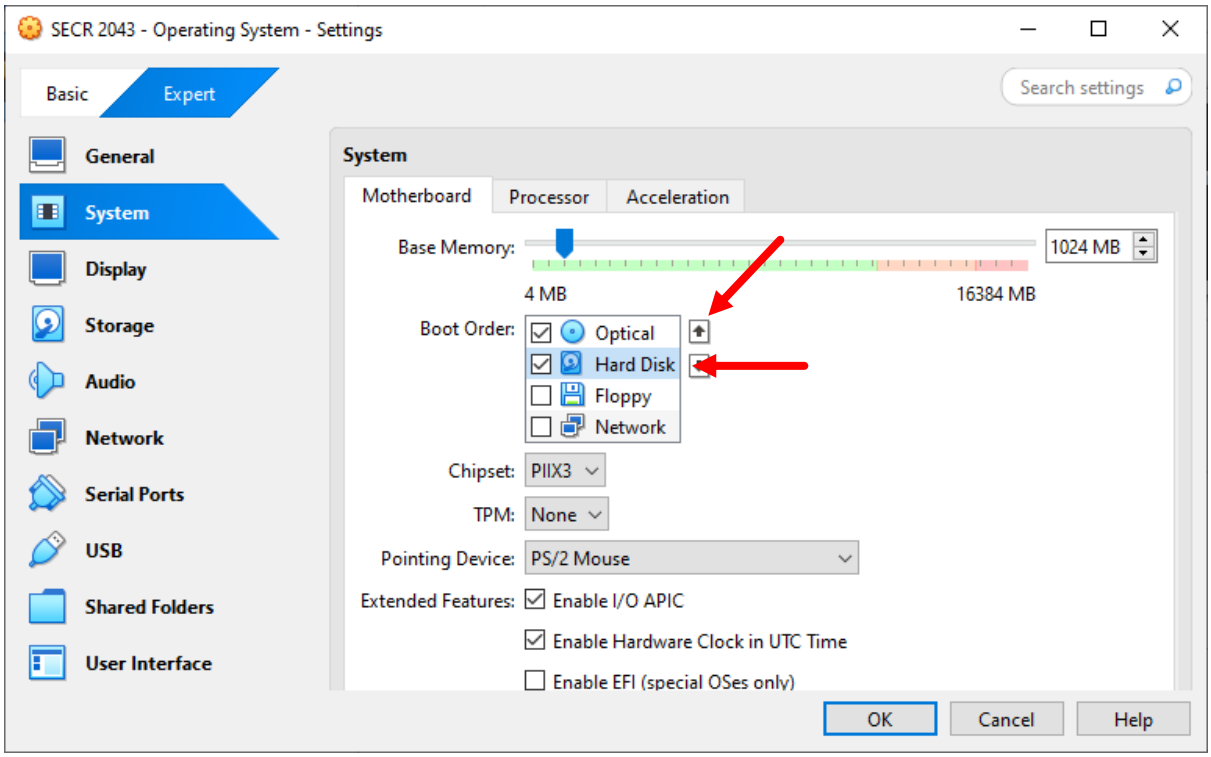
Turn off the Floppy disk boot option by unchecking the **Floppy** box, as shown



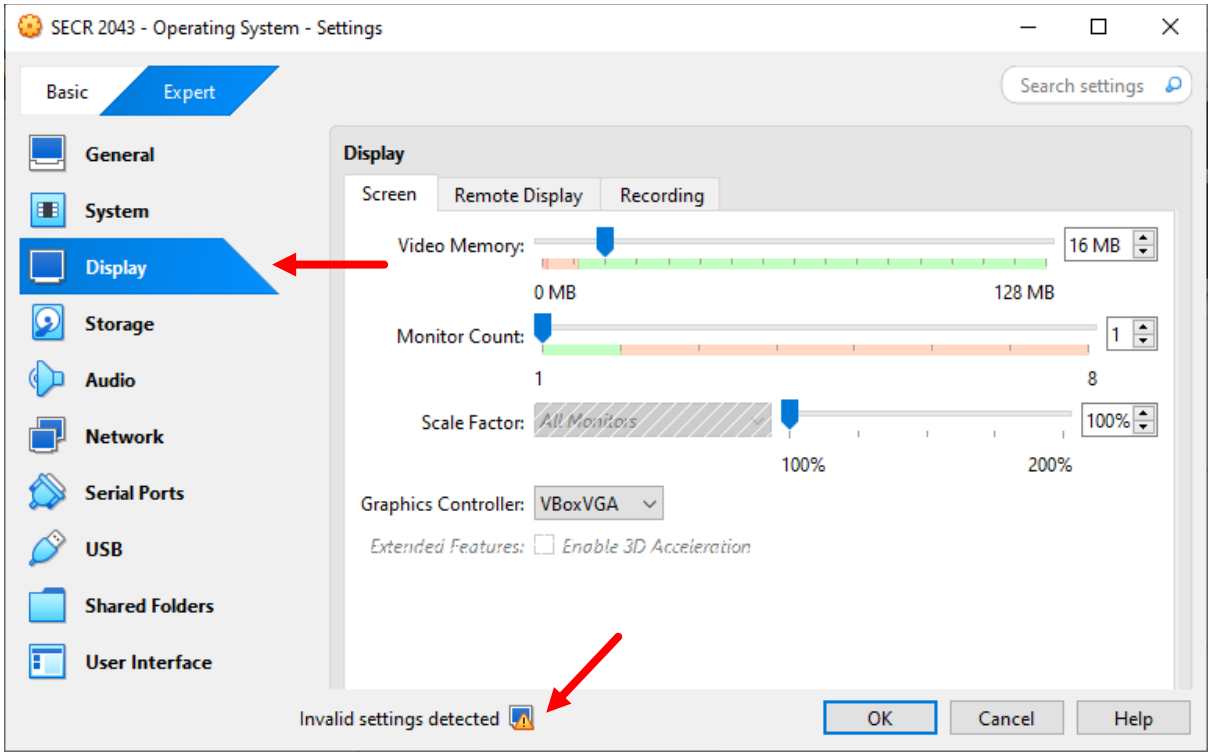
Next, select the **Optical** device and move it to the top of the boot order.



Do the same to **Hard Disk** as second boot device in **Boot Order**.



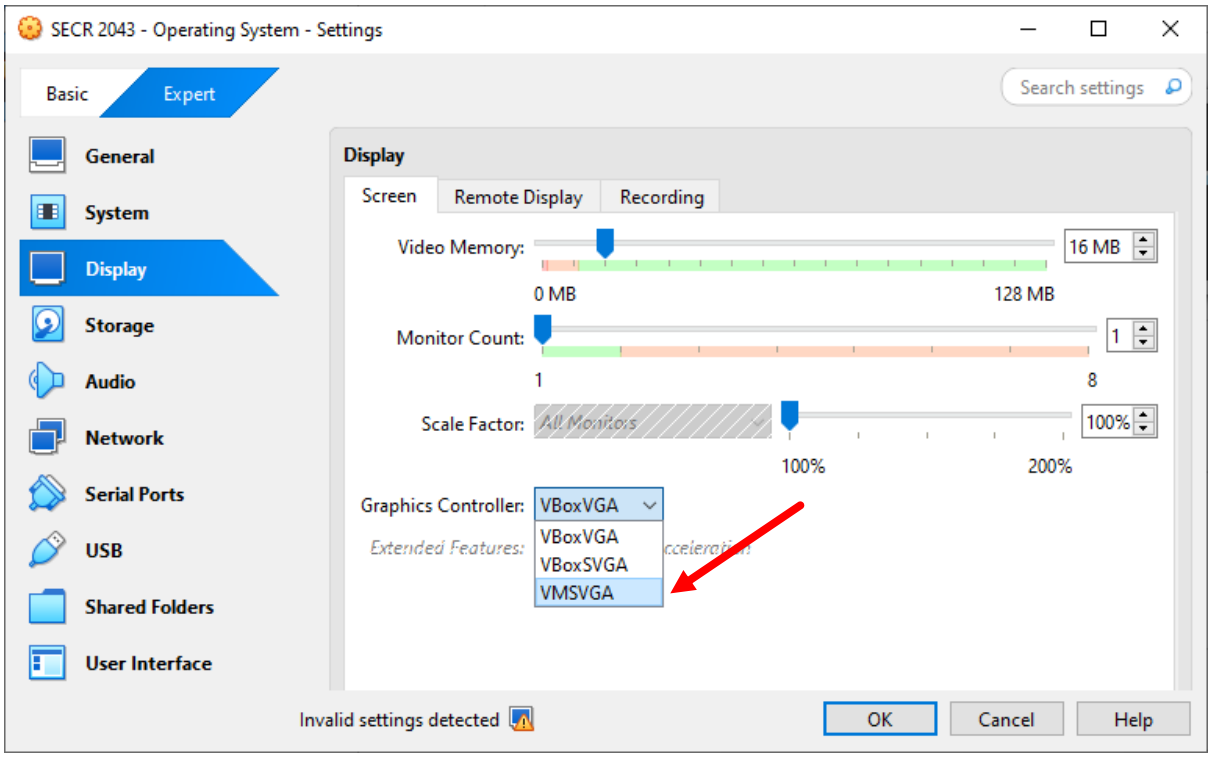
Now we move to Display setting of our virtual machine to fix some setting error as shown in the screenshot below.



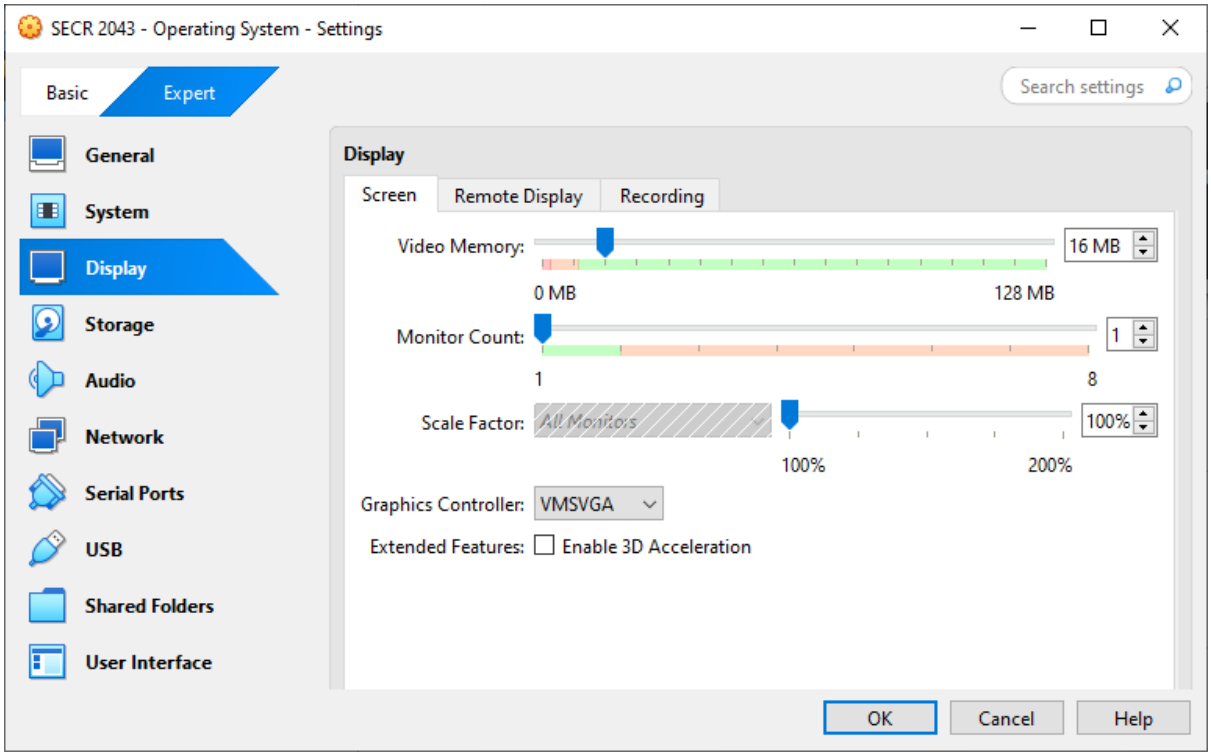
SCSR2043 - OPERATING SYSTEMS (Lab Tutorial 0)

Written by: Dr. Farkhana Muchtar

Change the **Graphics Controller** from **VBoxVGA** to **VMSVGA** like below.



Now, we don't have invalid setting warning message anymore.



Don't forget to click OK button to confirm the configuration changes.

SCSR2043 - OPERATING SYSTEMS (Lab Tutorial 0)

Written by: Dr. Farkhana Muchtar

Step 4: seed.iso for appliance setup

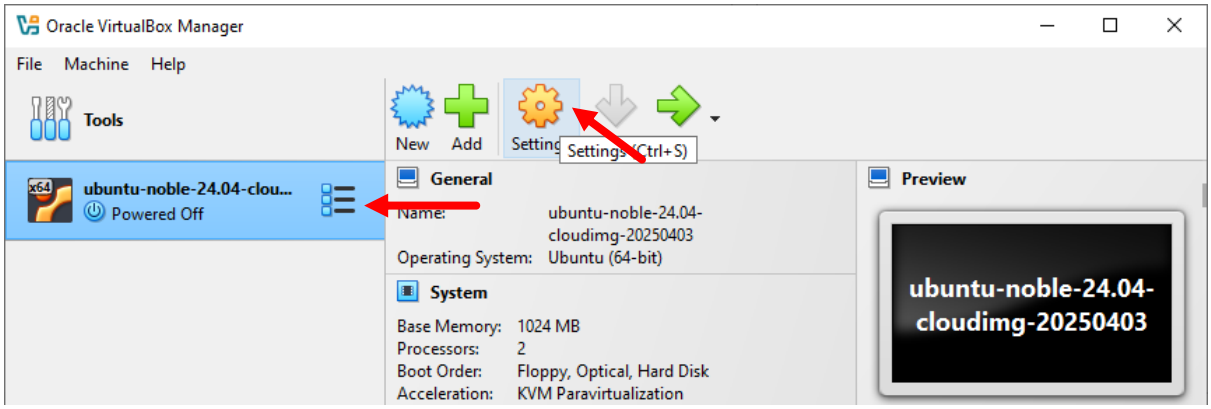
By default, you want to be able to log in to this minimal Ubuntu server image but there is no existing user with password.

To fix this, we'll use a **seed.iso** file to set up a username and password. We'll also use the **seed.iso** file to pre-set the network, hostname, and basic SSH server settings so you can remotely access the virtual machine.

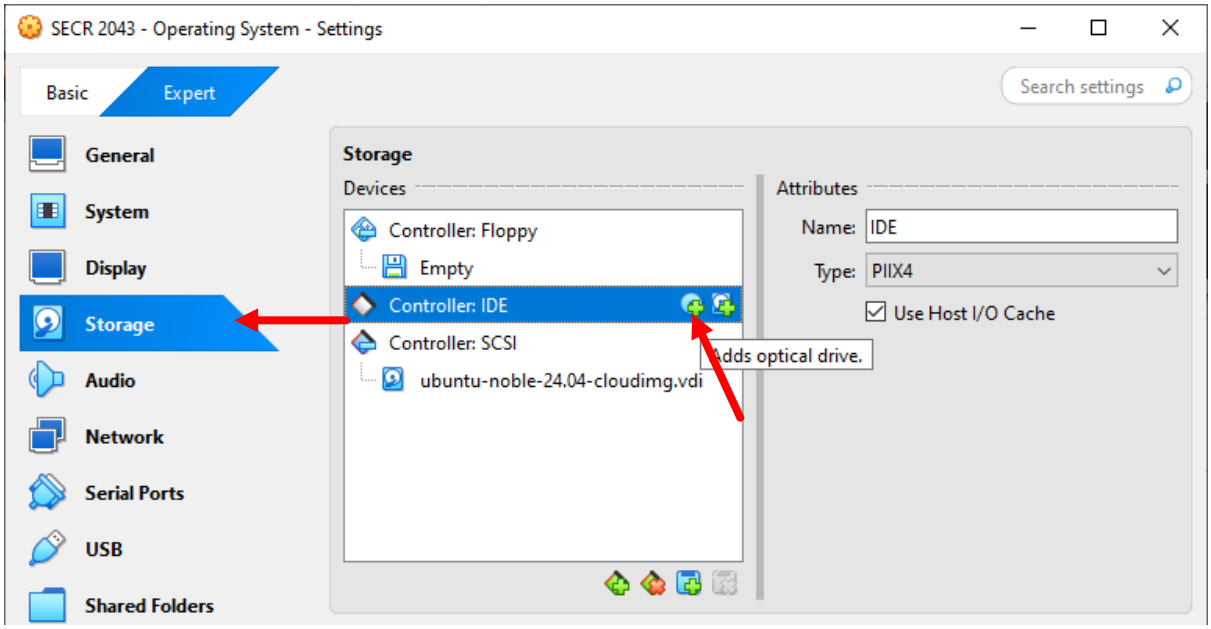
First of all, you need to download **seed.iso** that I created for this lab exercise from the link below and save it at any location that you desired such as Downloads folder.

https://drive.google.com/uc?export=download&id=1RyGd4MwSDMT_TglIsdX9jQoPSlyNxFj3

Now, select the **imported** Virtual machine (created using the Cloud image OVA) and hit the **Settings** button.

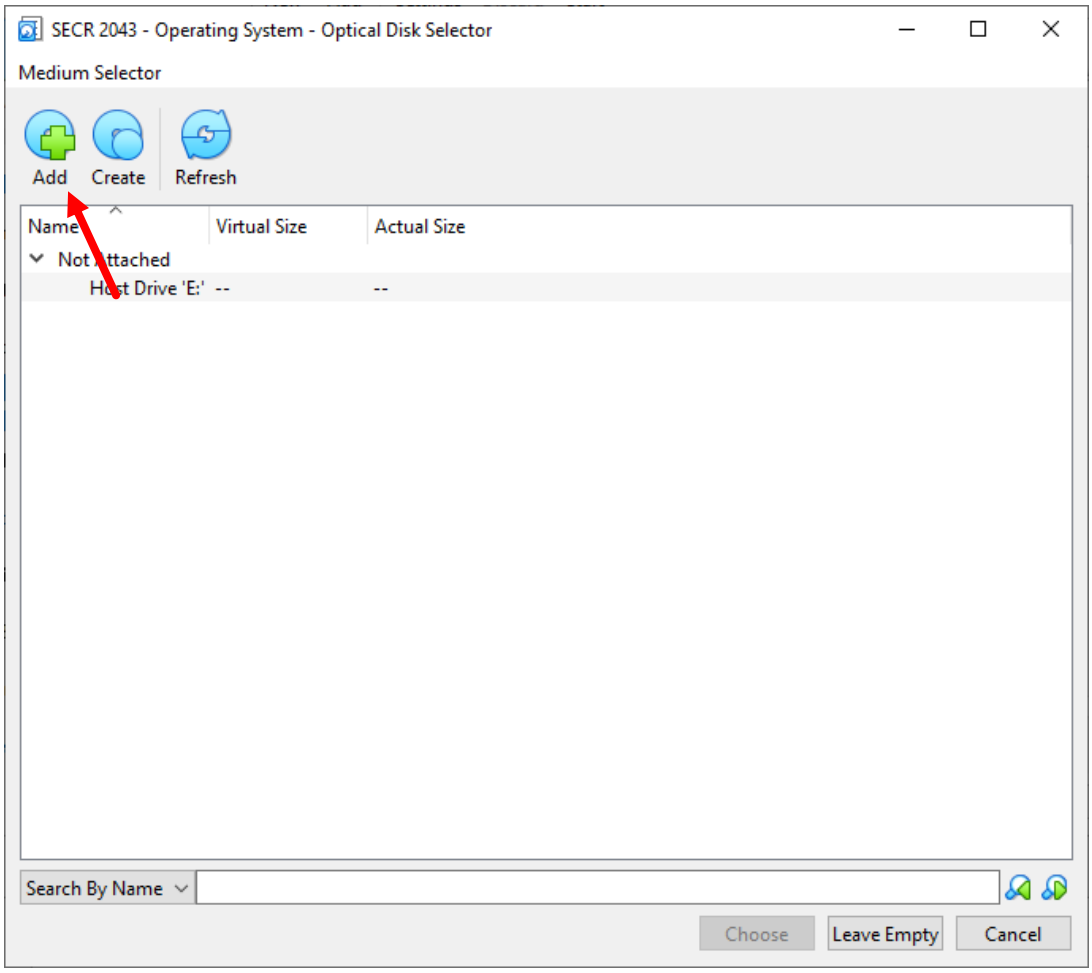


After that, click to **Storage**, and then click **add optical drive** icon at **Controller IDE** to add **seed.iso** as new virtual cd to our virtual machine.

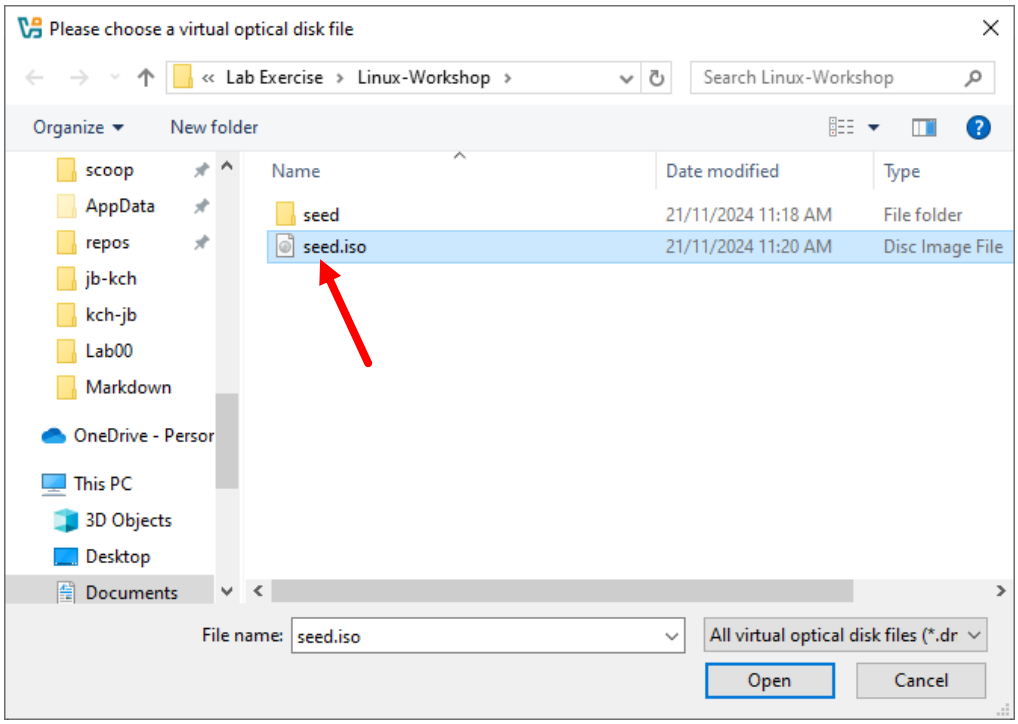


Written by: Dr. Farkhana Muchtar

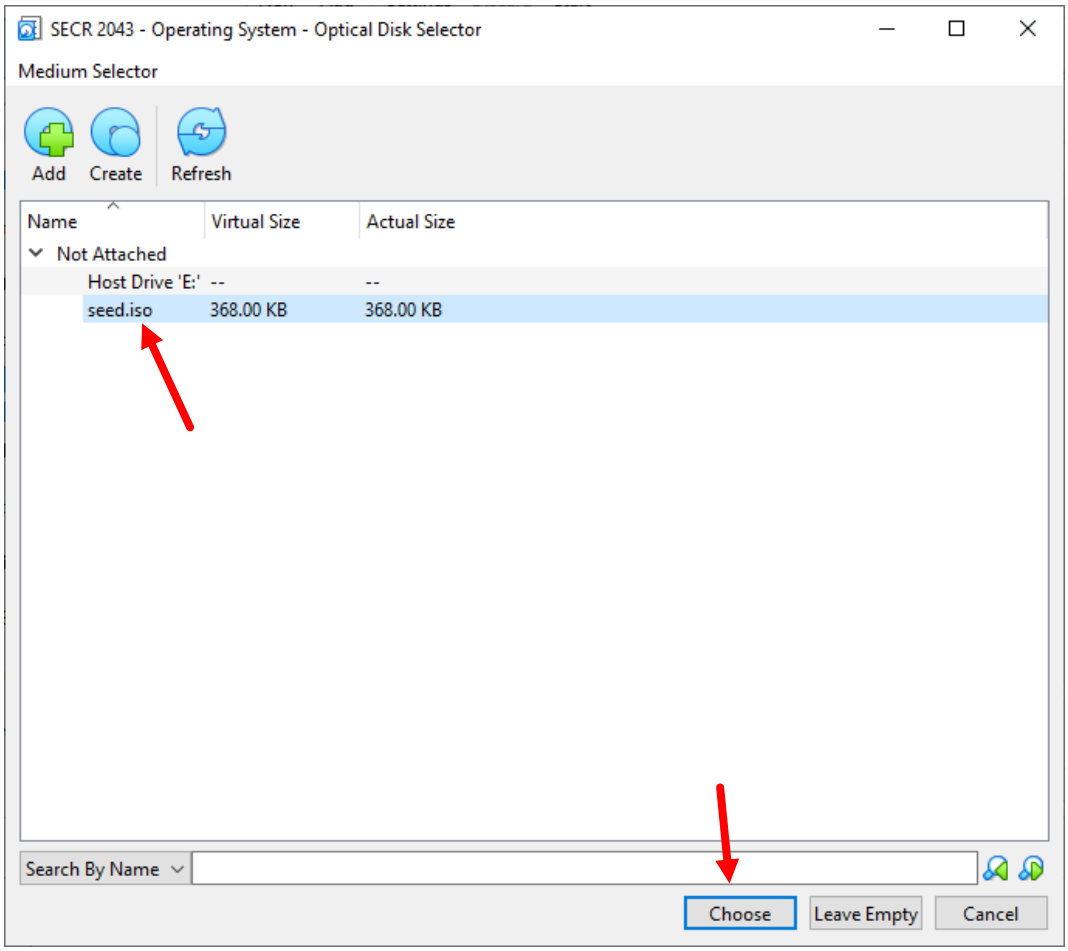
To do that, click on **Add** button and file explorer will open.



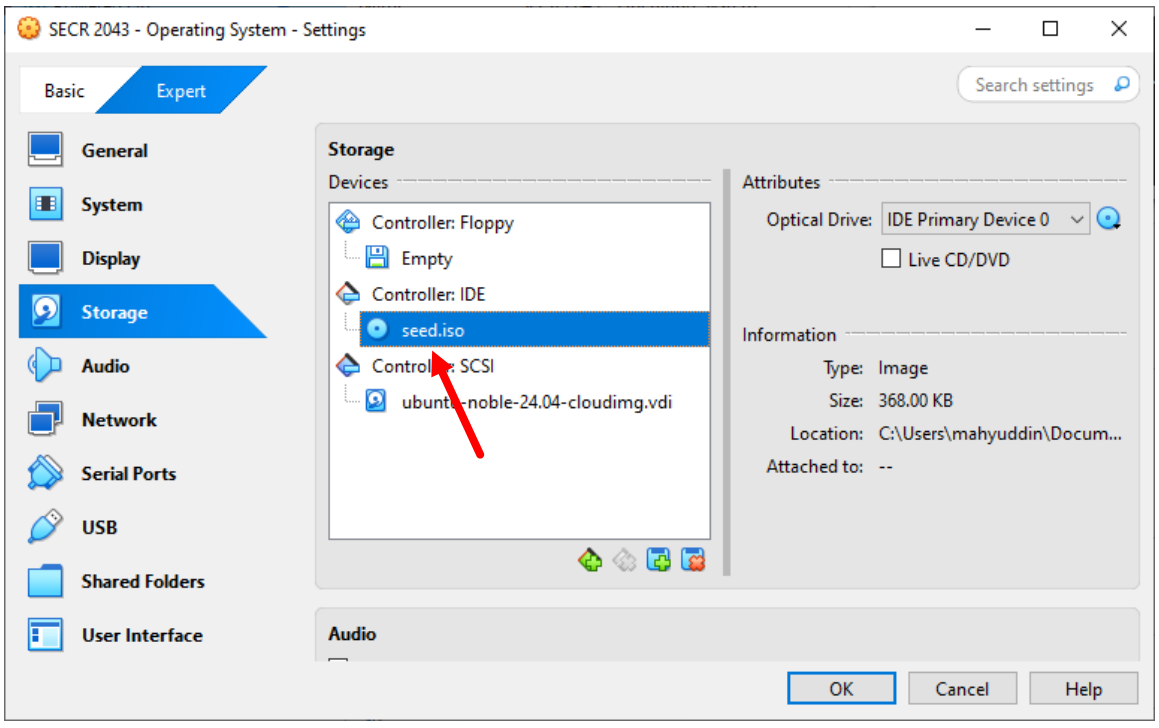
Now, select the **seed.iso** file which you have downloaded.



After that, it will show in the **File selection** area, **again select** it and hit the **Choose** button.



Now, you have **seed.iso** as bootable virtual optical disk (CD) for you virtual machine



SCSR2043 - OPERATING SYSTEMS (Lab Tutorial 0)

Written by: Dr. Farkhana Muchtar

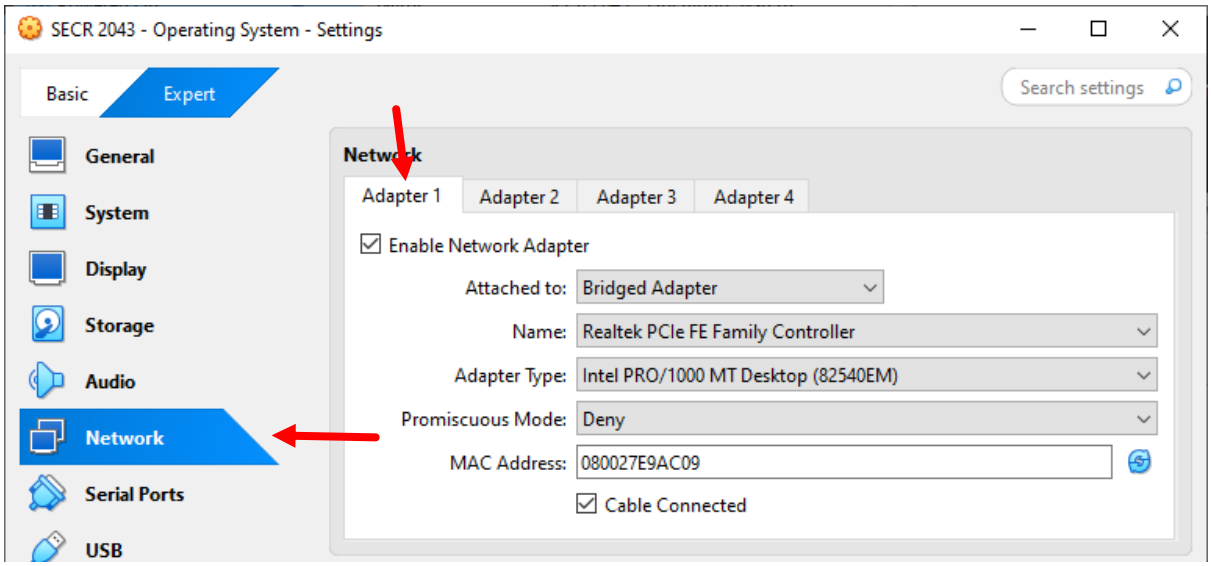
Step 5: Network Adapter Setup

We need to set up two network adapters for this virtual machine.

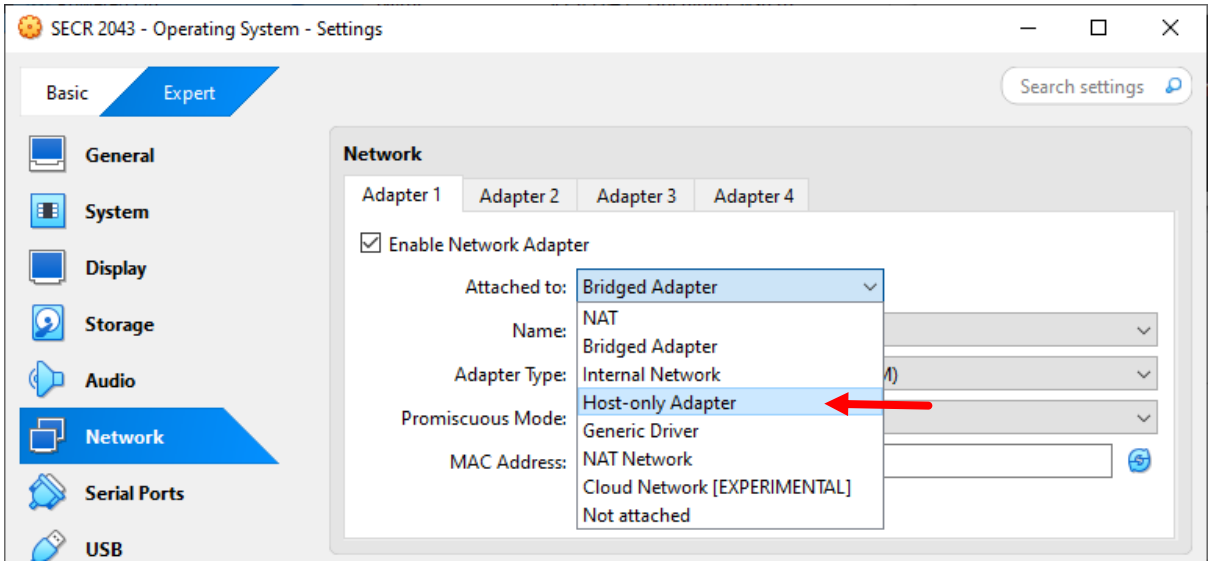
1. **Adapter 1** lets us connect to the virtual machine remotely using SSH.
2. **Adapter 2** is for this virtual machine to have Internet connection.

To achieve that, we set **Adapter 1** using **Host-Only Adapter**

Originally, Adapter 1 is attached to **Bridged Adapter**.

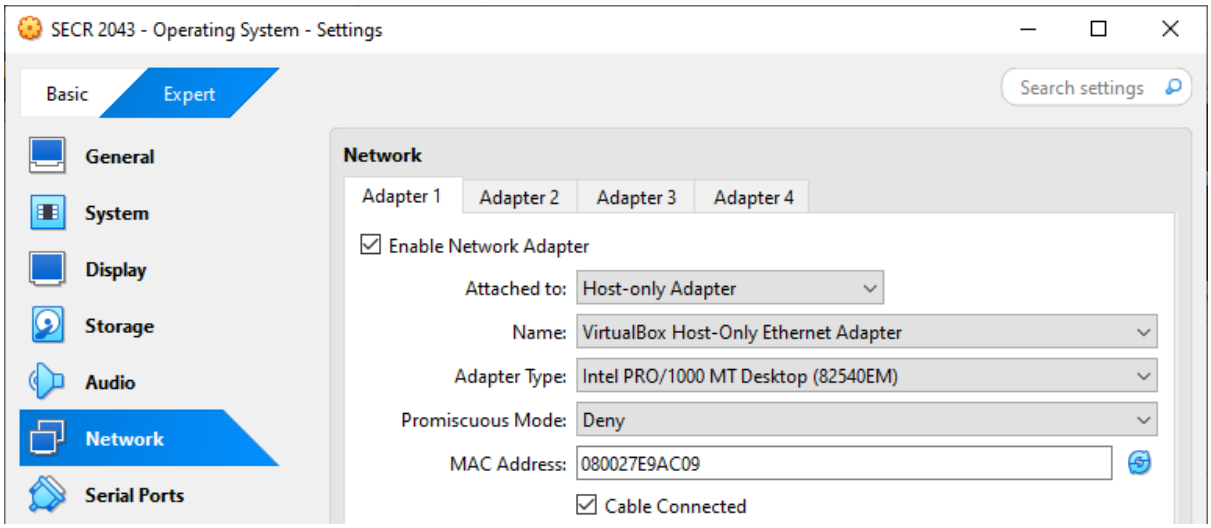


Change the **Bridged Adapter** to **Host-only Adapter** for **Adapter 1**.

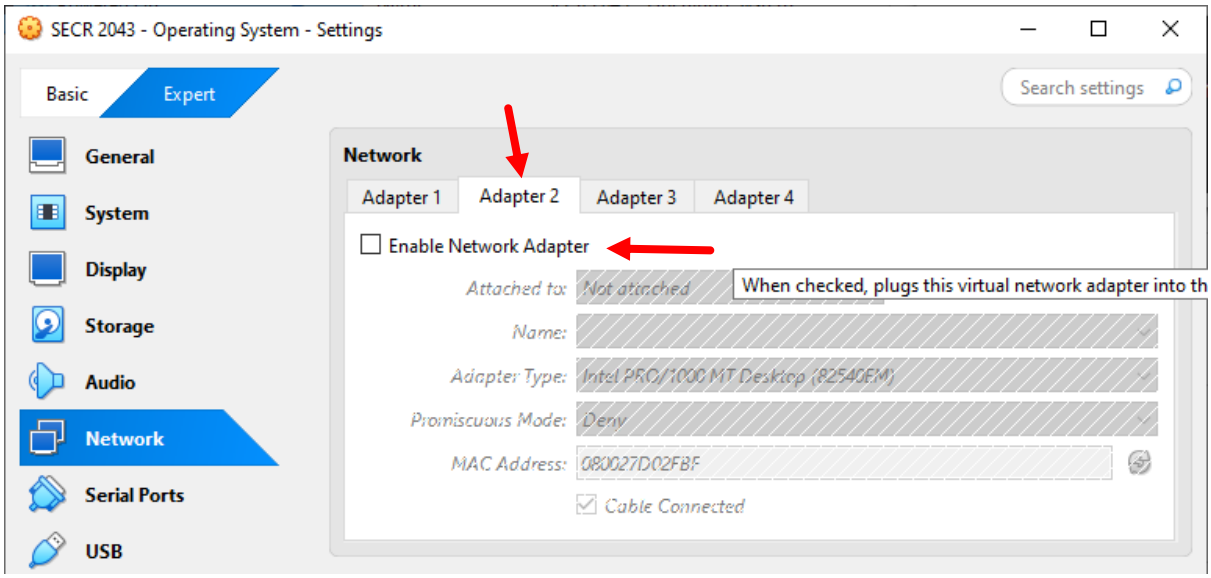


Written by: Dr. Farkhana Muchtar

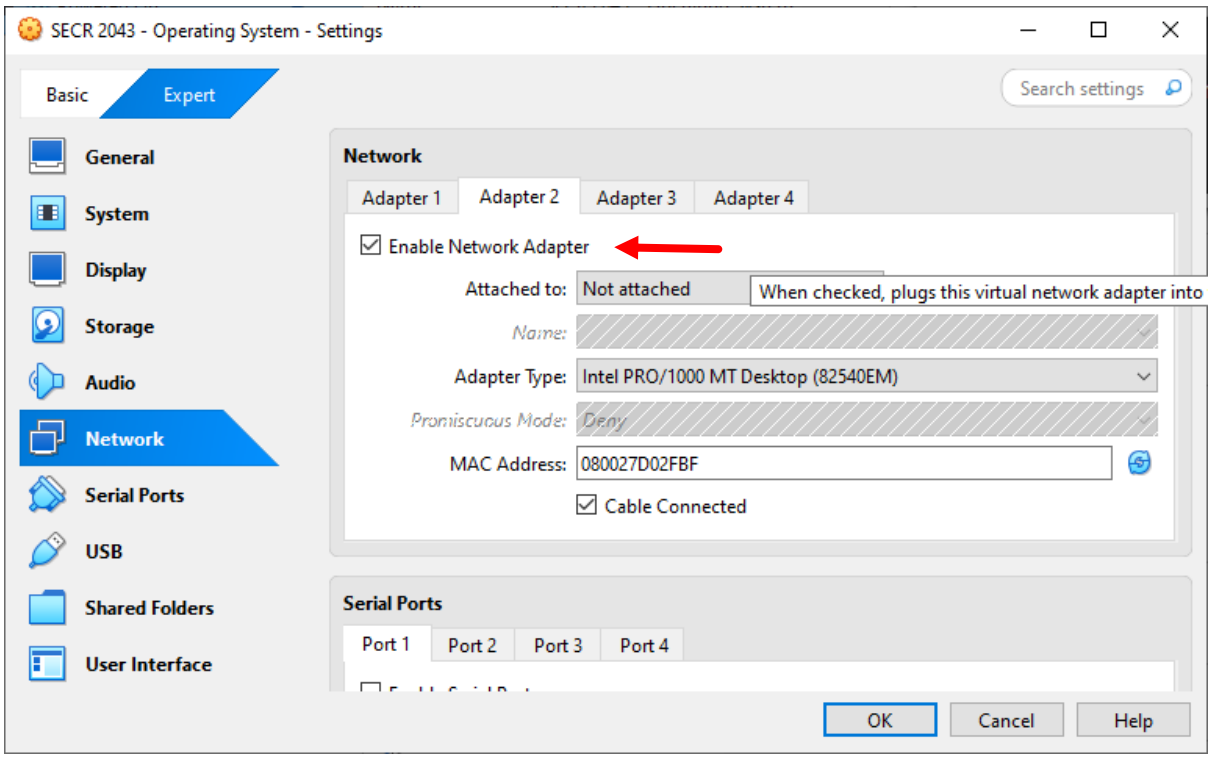
Now we have **Adapter 1** that is attached to **Host-only Adapter**. This adapter is dedicated to allow host machine (your laptop) to access the virtual machine remotely.



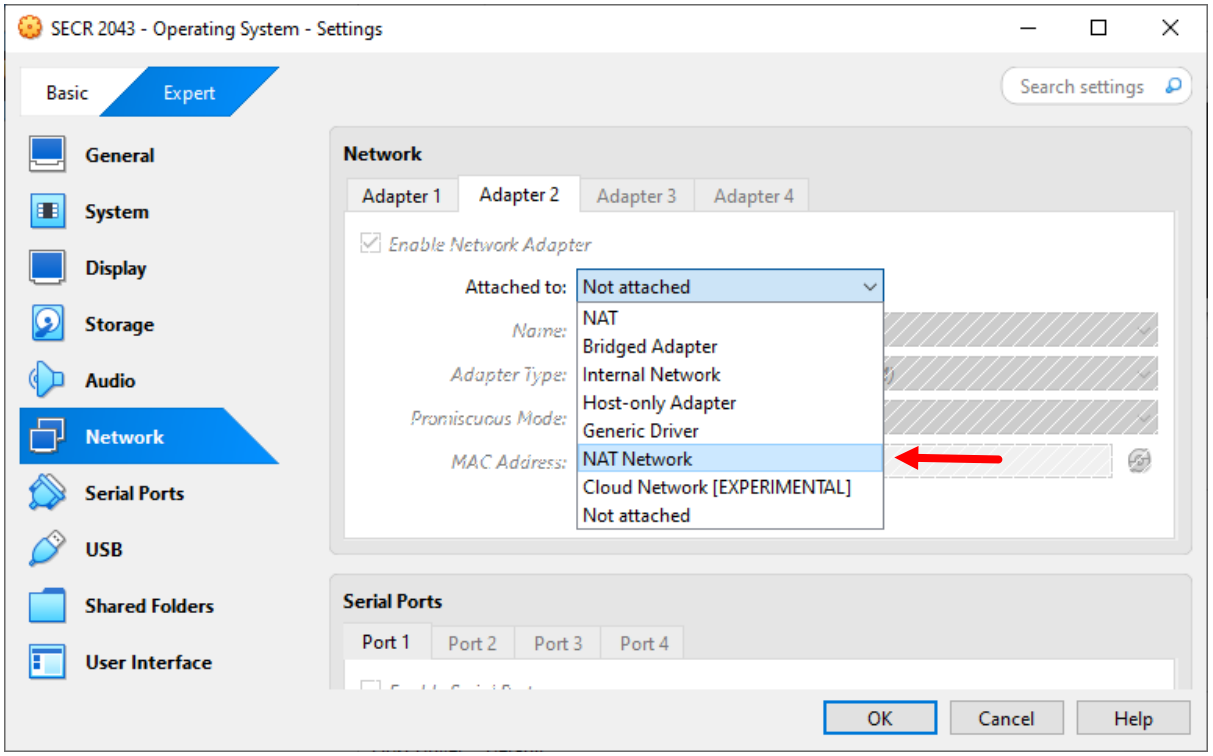
Next, we move to the **Adapter 2**. By default, this adapter is **disabled**. Just enable it by checking the **Enable Network Adapter**.



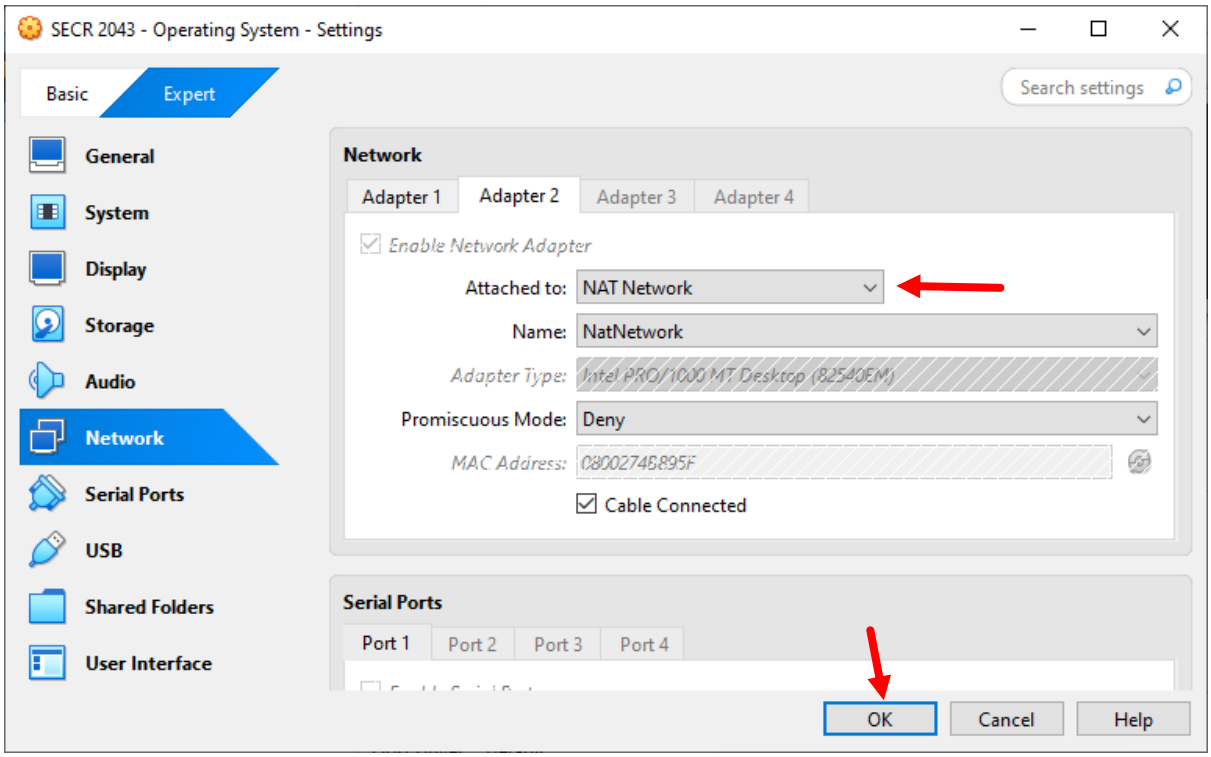
Written by: Dr. Farkhana Muchtar



Next is to change the Attached to from **Not attached** to **NAT Network**.

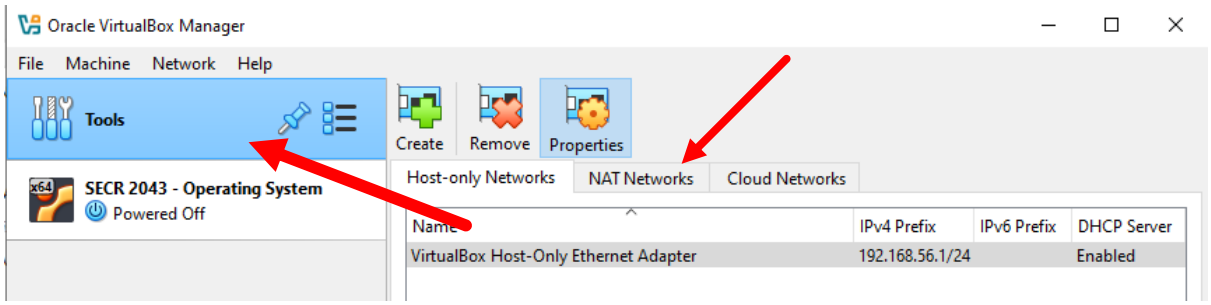


Don't forget to click **OK** button to confirm the changes.

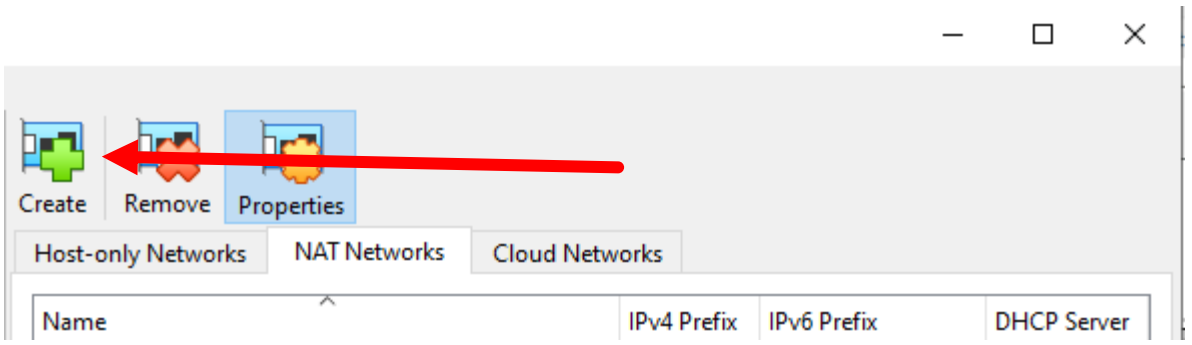


Notes:

If there is no NAT network available like above, you might have to add NAT network manually. Go to Tools (as shown below), and choose NAT Network.

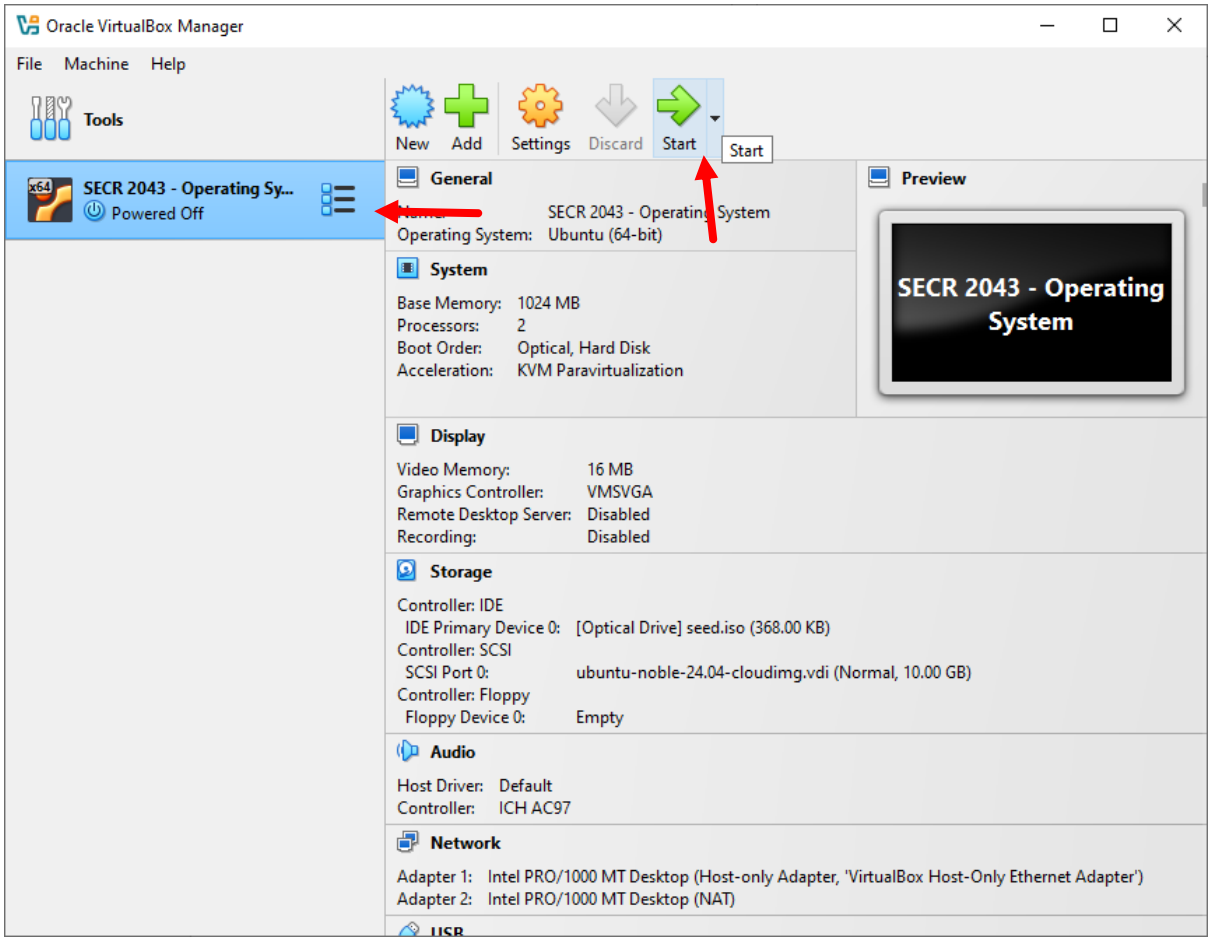


Add new NAT Network by clicking Create button.



Written by: Dr. Farkhana Muchtar

Start the virtual machine for the first time by selecting the created virtual machine and click Start icon.



Written by: Dr. Farkhana Muchtar

1.4 Post Installation

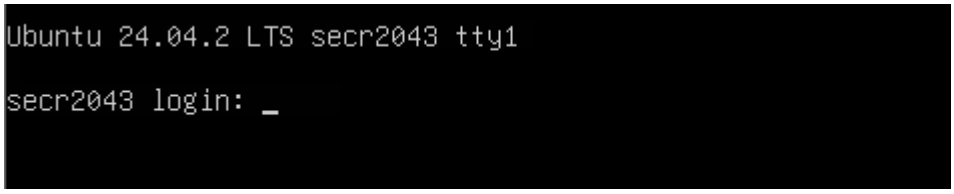
1.4.1 Login and Change password

Step 1: Login to the virtual machine
By default, the username is 'student' and the password is 'os123'.

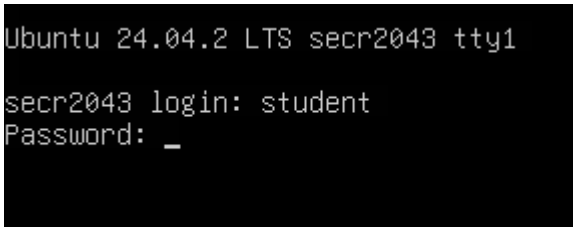
Username = **student**
Password = **os123**

Enter the same to log in to your **Ubuntu 24.04 minimal Cloud Image** running on VirtualBox.

When the login screen appear like below. Just type the username next to the login prompt and enter.



Next, it will ask you to enter the password of the current user.



When you type the password, it won't show on the screen. Don't worry, this is normal.

This is the classic security feature of any Unix or Unix like operating system when you accessing the machine remotely using ssh, by **not displaying the typed password**.

1.4.2 Add a new user with sudo access

Step 1: Well, we don't want to use default user, so create a **non-root user** with **sudo** rights, using the given commands:

```
sudo addduser muhammad
```

You can replace **muhammad** with whatever user name you want to give your new user.

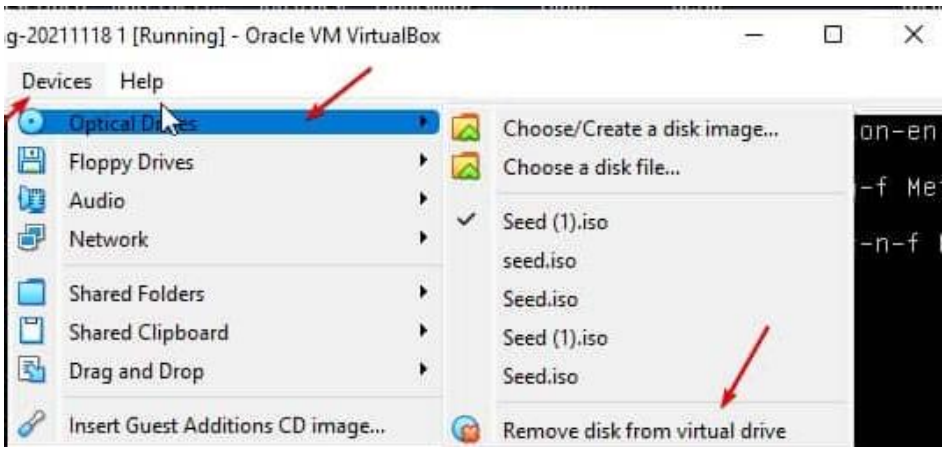
Add created user to the sudo group.

```
sudo usermod -aG sudo muhammad
```

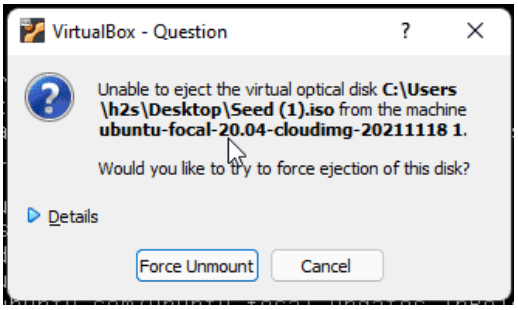
sudo is shortform of "super user do". Any user that is part of sudo group, has the authority of super user such as software installation, system configuration and deleting system files.

1.4.3 Remove Seed ISO and restart your system

Step 1: Go to **Devices**-> Select **Optical Devices** and then **Remove disk from virtual drive**.



Step 2: After that click **Force Unmount** button.



Step 3: Shutdown the virtual machine

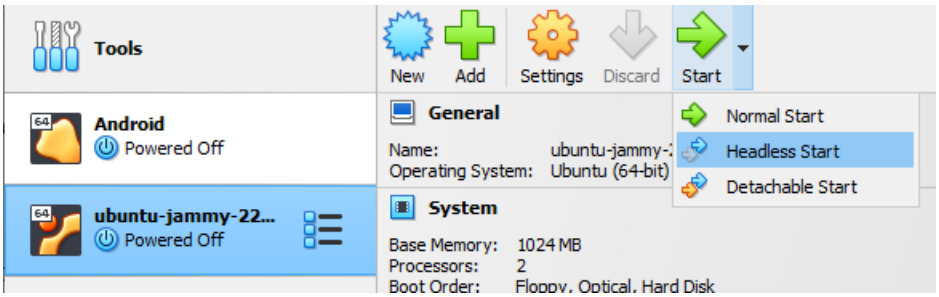
Now, we have removed the **seed.iso** file, simply shutdown your virtual machine system using this command:

```
sudo halt -p
```

Written by: Dr. Farkhana Muchtar

2 Accessing Ubuntu 24.04 Virtual Machine Remotely

Step 1: Run Ubuntu Server 24.04 with headless start mode.



Step 2: Open Windows Terminal

Step 3: Run this command

```
ssh <username>@192.168.56.10
```

Use new username that you have created at section 1.4.2
e.g:-

```
ssh muhammad@192.168.56.10
```

The first time you use SSH, it will ask to add a public key for security. Just type **yes** to confirm.

As before, **your password won't appear** when you type it—this is a security feature.

```
~ $ ssh muhammad@192.168.56.10
The authenticity of host '192.168.56.10 (192.168.56.10)' can't be established.
ED25519 key fingerprint is SHA256:5Q819UGtKK2jgQPEBLtt1EZG2XqgX9+G3Os9b2MesWI.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.10' (ED25519) to the list of known hosts.
muhammad@192.168.56.10's password:
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

muhammad@secr2043:~$
```

Done.